Systemadministration

Introduktion

En introduktion till datorer

Förberedelse

Läshänvisningar (vecka 1)

Veckans kurslitteratur finns tillgängligt som PDF-utdrag eller online genom en extern webbplattform.

Operating System Concepts 10th ed. [sammanlagt 17 sidor]

Silberschatz, A., Galvin, P. B., & Gagne, G., 2019

- **1.1 What Operating Systems Do**, sida 3-7
- **1.4.1 Multiprogramming and Multitasking**, sida 23-24
- **1.7 Virtualization**, sida 34-35
- **2.1 Operating-System Services** & **2.2 User and Operating-System Interface**, sida 55-61
- **2.4 System Services**, sida 74-75

Security in Computing 5th ed. [sammanlagt 13 sidor]

Pfleeger, C. P., Pfleeger, S. L., & Margulies, J., 2015

- **5. Operating Systems**, sida 280-289
- **5. Operating Systems**, sida 312-314

MD-100 Support the Windows client environment [4 moduler, 1 h 32 min]

Microsoft Learn https://learn.microsoft.com/en-us/training/paths/support-windows-client-environment/

https://ju.instructure.com/courses/9876/pages/lashanvisningar-vecka-1?module_item_id=282666

Agenda

En överblick över hur en dator fungerar

1. Computers	Datorer
2. Programs	Program
3. Software	Mjukvara
4. Operating Systems	Operativsystem
5. Hardware	Hårdvara

1. Computers

Vad är en dator?

Vilken av följande är en dator?

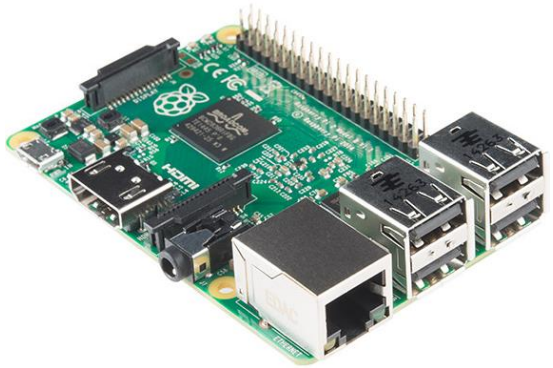
iPhone, by Jared Earle, 2010, Flickr (http://flic.kr/p/7L37A1). CC BY-NC-ND 2.0.

Apple Watch, by Tim Reckmann, 2017, Flickr (http://flic.kr/p/228EqFf). CC BY 2.0.

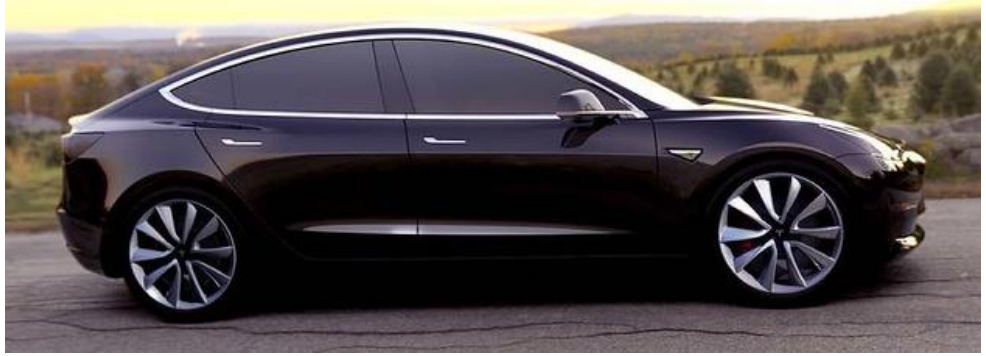
01_Ideapad_530S_Hero_Presentation_mode_OnyxBlack_B_NG, by BTNHD Production, 2018, Flickr (http://flic.kr/p/25DqcK2). CC BY-NC-ND 2.0.

Vad är en dator?

Vilken av följande är en dator?



Raspberry Pi 2 - Model B, by SparkFun Electronics, 2015, Flickr (http://flic.kr/p/rsueQP). CC BY 2.0.



215-Mile Range "Tesla Model 3" Electrifies the World, by Automotive Rhythms, 2016, Flickr (http://flic.kr/p/EX2Eud). CC BY-NC-ND 2.0.

Frontansicht_oben_AntennenW, by Timo Schmitt, 2016, Flickr (http://flic.kr/p/GuU7fP). CC BY-NC 2.0.

What is a Computer?

WIKIPEDIA
The Free Encyclopedia

Computer (occupation)

The term "**computer**", in use from the early 17th century (the first known written reference dates from 1613), meant "one who computes": a person performing mathematical calculations.

Wikipedia contributors. (2023, July 22). Computer (occupation). In *Wikipedia, The Free Encyclopedia*. Retrieved 08:43, July 29, 2023, from https://en.wikipedia.org/w/index.php?title=Computer_(occupation)&oldid=1166531475

Woman Computer with Microscope and Calculator [L-74768], by NACA (now NASA), 1952, Flickr (http://flic.kr/p/fqCRSL). No restrictions.

What is a Computer?

WIKIPEDIA
The Free Encyclopedia

Computer

A **computer** is a machine that can be programmed to carry out sequences of arithmetic or logical operations (computation) automatically. Modern digital electronic computers can perform generic sets of operations known as programs. These programs enable computers to perform a wide range of tasks.

Wikipedia contributors. (2023, July 27). Computer. In *Wikipedia, The Free Encyclopedia*. Retrieved 08:31, July 29, 2023, from https://en.wikipedia.org/w/index.php?title=Computer&oldid=1167445805

2. Programs

Begrepp

Program
Application
App

Definition

ComputerSweden FROM IDG

IT-ord

Ord och uttryck i it-branschen

program

komplett uppsättning instruktioner, med början och slut, som utför en eller flera uppgifter i en dator.

IDG Sverige. (2018, November 14). Program. In *IT-ord*. Retrieved 13:30, July 31, 2023, from https://it-ord.idg.se/ord/program/

Computer Programs

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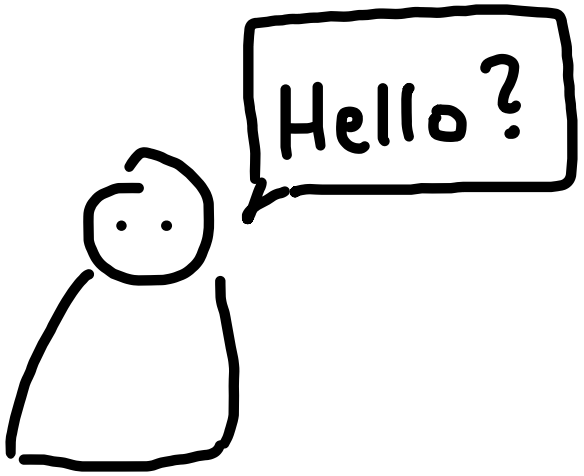
3 Software

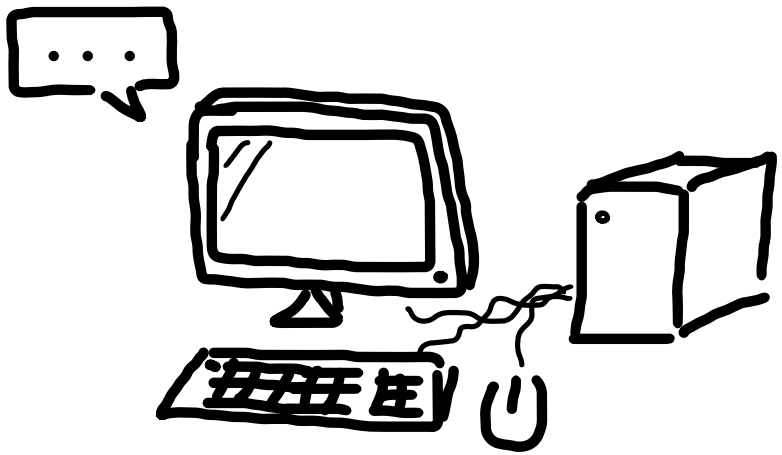
Software

Mjukvara

Datorer talar inte samma språk som människor…



Natural Language



Computer Language

Software

Mjukvara

Program är en samling instruktioner (**kod**)

Datorers ’modersmål’ kallas för **maskinspråk**

```
00000010  03 00 3e 00 01 00 00 00  f0 31 00 00 00 00 00 00
00000020  40 00 00 00 00 00 00 00  18 a2 00 00 00 00 00 00
00000030  00 00 00 00 40 00 38 00  0d 00 40 00 1e 00 1d 00
00000040  06 00 00 00 04 00 00 00  40 00 00 00 00 00 00 00
00000050  40 00 00 00 00 00 00 00  40 00 00 00 00 00 00 00
00000060  d8 02 00 00 00 00 00 00  d8 02 00 00 00 00 00 00
00000070  08 00 00 00 00 00 00 00  03 00 00 00 04 00 00 00
00000080  18 03 00 00 00 00 00 00  18 03 00 00 00 00 00 00
00000090  18 03 00 00 00 00 00 00  1c 00 00 00 00 00 00 00
000000a0  1c 00 00 00 00 00 00 00  01 00 00 00 00 00 00 00
```

```
mov     ah, 09h
lea     dx, msg
int     21h
mov     ax, 4c00h
int     21h
```

Software

Mjukvara

För att underlätta utveckling skrivs sällan program i **maskinkod**

Istället används ett **programmeringsspråk**

EX.

C	Javascript
Go	Python
Rust	Bash
Java	Perl

```c
static void
next_line_num (void)
{
  char *endp = line_num_end;
  do
    {
      if ((*endp)++ < '9')
        return;
      *endp-- = '0';
    }
  while (endp >= line_num_start);

  if (line_num_start > line_buf)
    *--line_num_start = '1';
```

GNU Coreutils contributors. (2023, July 31). cat.c. In *GNU Savannah*. Retrieved 07:25, August 1, 2023, from https://git.savannah.gnu.org/cgit/coreutils.git/tree/src/cat.c

Compiler

Kompilator

Ett program som översätter **källkod/source code** till **maskinkod**

```
132 static void
133 next_line_num (void)
134 {
135   char *endp = line_num_end;
136   do
137     {
138       if ((*endp)++ < '9')
139         return;
140       *endp-- = '0';
141     }
142   while (endp >= line_num_start);
143
144   if (line_num_start > line_buf)
145     *--line_num_start = '1';
146   else
147     *line_buf = '>';
148   if (line_num_start < line_num_print)
```

```
03 00 3e 00 01 00 00 00  f0 31 00 00 00 00 00 00
40 00 00 00 00 00 00 00  18 a2 00 00 00 00 00 00
00 00 00 00 40 00 38 00  0d 00 40 00 1e 00 1d 00
06 00 00 00 04 00 00 00  40 00 00 00 00 00 00 00
40 00 00 00 00 00 00 00  40 00 00 00 00 00 00 00
d8 02 00 00 00 00 00 00  d8 02 00 00 00 00 00 00
08 00 00 00 00 00 00 00  03 00 00 00 04 00 00 00
18 03 00 00 00 00 00 00  18 03 00 00 00 00 00 00
18 03 00 00 00 00 00 00  1c 00 00 00 00 00 00 00
1c 00 00 00 00 00 00 00  01 00 00 00 00 00 00 00
01 00 00 00 04 00 00 00  00 00 00 00 00 00 00 00
```

GNU Coreutils contributors. (2023, July 31). cat.c. In *GNU Savannah*. Retrieved 07:25, August 1, 2023, from https://git.savannah.gnu.org/cgit/coreutils.git/tree/src/cat.c

Interpreter

Interpretator

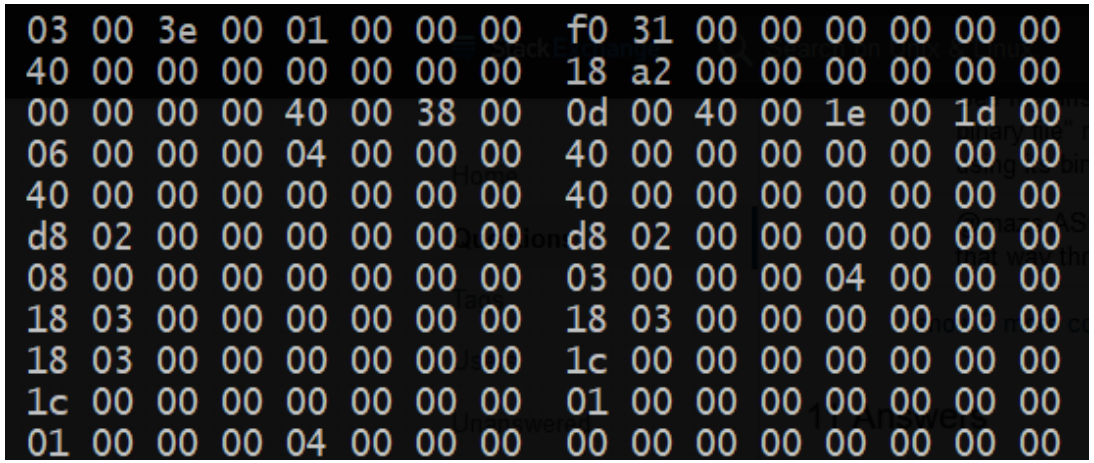
Ett program som översätter* **källkod/source code** stegvis medans programmet körs

```python
n = int(input('Type a number, and its factorial will be print

if n < 0:
    raise ValueError('You must enter a non-negative integer')

factorial = 1
for i in range(2, n + 1):
    factorial *= i

print(factorial)
```

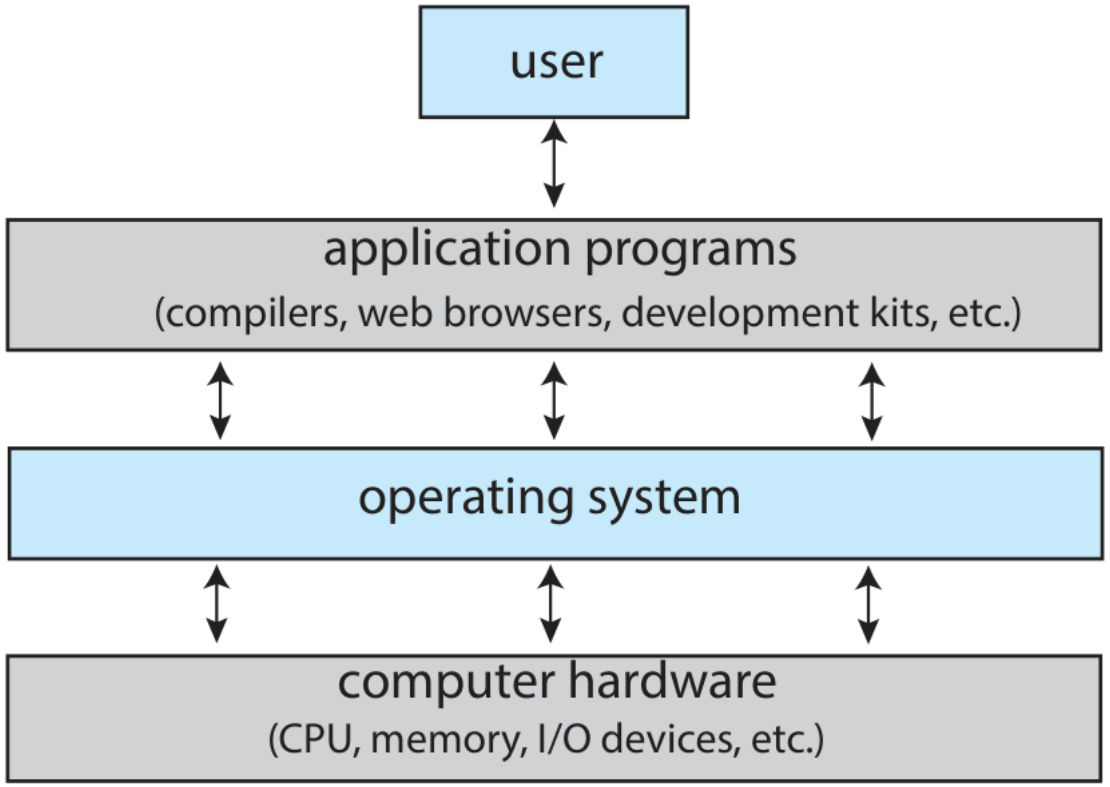


Wikipedia contributors. (2023, July 31). Python (programming language). In *Wikipedia, The Free Encyclopedia*. Retrieved 07:25, August 1, 2023, from https://en.wikipedia.org/w/index.php?title=Python_(programming_language)&oldid=1168084792

*förenklad förklaring

4. Operating Systems

Abstract view of the components of a computer system



From *Operating System Concepts* (10th edition), by A. Silberschatz, P. B. Galvin, G. Gagne, 2019, Wiley.
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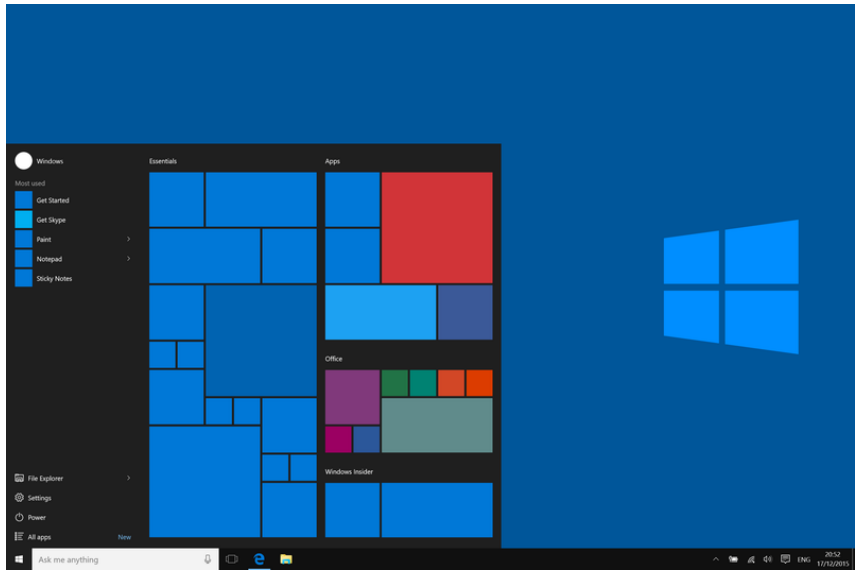
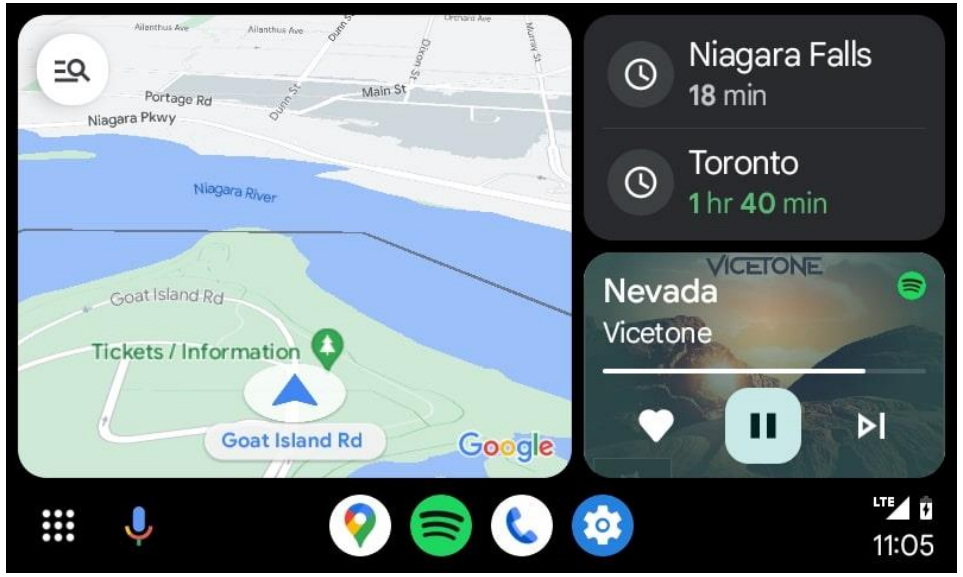
Operating Systems
Operativsystem



Abstract version of Windows 10 interface, by Microsoft, 2015, Wikimedia Commons (https://commons.wikimedia.org/w/index.php?curid=45697510). No license, trademark of Microsoft Corporation.

Android 11 Homescreen, by Amy.1731, 2021, Wikimedia Commons (https://commons.wikimedia.org/w/index.php?curid=100483995). Apache 2.0.



Visuel de l'écran d'accueil Android Auto, by Paowee, 2023, Wikimedia Commons (https://commons.wikimedia.org/w/index.php?curid=36248043). CC BY-SA 4.0.

Operating Systems

Operativsystem

En Liknelse…

- Säkerhetskontroll
- Flygledartorn
- Flygplan
- Flygbana

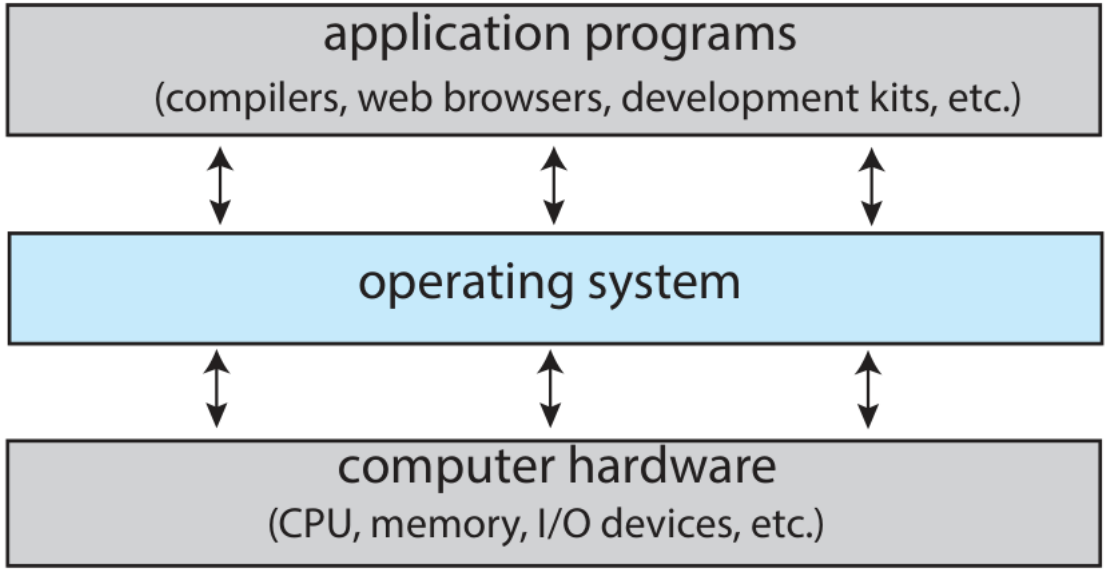
Aerial view of airport with lots of airplanes during daytime, by skyler_tv, 2016, Unsplash (https://unsplash.com/photos/9xdho4stJQ8). Unsplash license.

Operating Systems

Operativsystem

En Liknelse…

- Säkerhetskontroll
- Flygledartorn
- Flygplan
- Flygbana



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Operating Systems

Operativsystem

Tillhandahåller funktioner för bl.a:

- Ett användargränssnitt
- Exekvering av program
- Filsystem
- Kommunikation mellan processer
- Tillgång till hårdvara
- Säkerhet

5. Hardware

iPhone, by Jared Earle, 2010, Flickr (http://flic.kr/p/7L37A1). CC BY-NC-ND 2.0.

Apple Watch, by Tim Reckmann, 2017, Flickr (http://flic.kr/p/228EqFf). CC BY 2.0.

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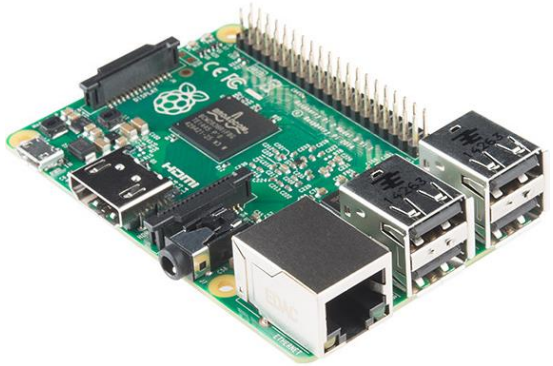
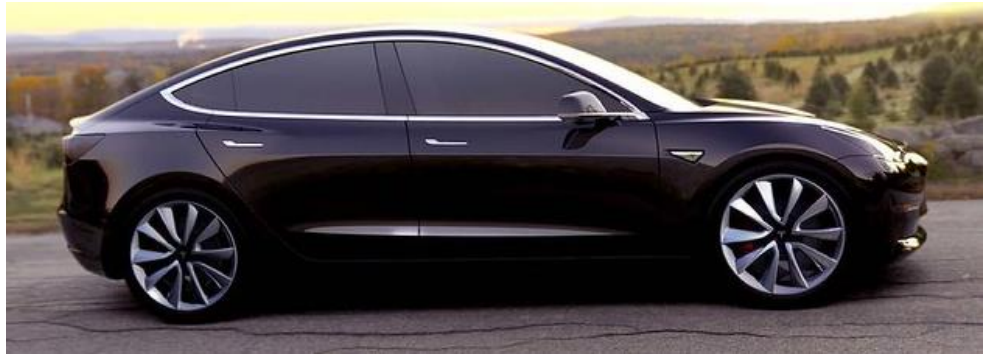


Raspberry Pi 2 - Model B, by SparkFun Electronics, 2015, Flickr (http://flic.kr/p/rsueQP). CC BY 2.0.



215-Mile Range "Tesla Model 3" Electrifies the World, by Automotive Rhythms, 2016, Flickr (http://flic.kr/p/EX2Eud). CC BY-NC-ND 2.0.

Frontansicht_oben_AntennenW, by Timo Schmitt, 2016, Flickr (http://flic.kr/p/GuU7fP). CC BY-NC 2.0.

Hardware

Hårdvara

MT36JSF1G72PZ-1G9K1HE; Memory Type DDR3 SDRAM; Capacity 8GB; Data Transfer Rate 1866MHz; Pins 240; Bus Type PC-14900; Error Correction Registered ECC; Cycle Time 1.07ns; Cas CL13; Memory Clock 233MHz; Voltage 1.5V; Manufacturer Micron Technology, Inc., by Mister rf, 2023, Wikimedia Commons (https://commons.wikimedia.org/w/index.php?curid=129056892). CC BY-SA 4.0

Primary Storage
Main Memory
Internal Memory
Core Memory

'Memory'

Top view of an Intel central processing unit Celeron Kaby Lake type core, model G3930, launched in 2017. LGA 1151 socket, 14 nm process, core frequency 2.90 GHz, Thermal Design Power 51 W, dimensions 37.5mm x 37.5mm. Manufactured in Vietnam. by Erica Gaba – Wikimedia Commons user: Sting, 2022, Wikimedia Commons (https://commons.wikimedia.org/w/index.php?curid=123062267). CC BY-SA 4.0

Processor

Secondary Storage
Persistent Storage
External Memory

'Storage'

'Disk'

Hårdvara

Hardware

MT36JSF1G72PZ-1G9K1HE; Memory Type DDR3 SDRAM; Capacity 8GB; Data Transfer Rate 1866MHz; Pins 240; Bus Type PC-14900; Error Correction Registered ECC; Cycle Time 1.07ns; Cas CL13; Memory Clock 233MHz; Voltage 1.5V; Manufacturer Micron Technology, Inc., by Mister rf, 2023, Wikimedia Commons (https://commons.wikimedia.org/w/index.php?curid=129056892). CC BY-SA 4.0

Primärminne
Internminne

‘Minne’

Top view of an Intel central processing unit Celeron Kaby Lake type core, model G3930, launched in 2017. LGA 1151 socket, 14 nm process, core frequency 2.90 GHz, Thermal Design Power 51 W, dimensions 37.5mm x 37.5mm. Manufactured in Vietnam. by Erica Gaba – Wikimedia Commons user: Sting, 2022, Wikimedia Commons (https://commons.wikimedia.org/w/index.php?curid=123062267). CC BY-SA 4.0

Processor

Sekundärminne
Externminne

‘Lagring’

‘Disk’

Hardware
Hårdvara

Top view of an Intel central processing unit Celeron Kaby Lake type core, model G3930, launched in 2017. LGA 1151 socket, 14 nm process, core frequency 2.90 GHz, Thermal Design Power 51 W, dimensions 37.5mm x 37.5mm. Manufactured in Vietnam. by Erica Gaba – Wikimedia Commons user: Sting, 2022, Wikimedia Commons (https://commons.wikimedia.org/w/index.php?curid=123062267). CC BY-SA 4.0

Processor
Utför instruktioner/beräkningar
‘Hjärnan i en dator’

MT36JSF1G72PZ-1G9K1HE; Memory Type DDR3 SDRAM; Capacity 8GB; Data Transfer Rate 1866MHz; Pins 240; Bus Type PC-14900; Error Correction Registered ECC; Cycle Time 1.07ns; Cas CL13; Memory Clock 233MHz; Voltage 1.5V; Manufacturer Micron Technology, Inc., by Mister rf, 2023, Wikimedia Commons (https://commons.wikimedia.org/w/index.php?curid=129056892). CC BY-SA 4.0

Primärminne
Fungerar som ett ‘arbetsminne’

Sekundärminne
Fungerar som ett ‘långtidsminne’

Hardware

Hårdvara

En Liknelse…

- Skrivbord
  Arbetsyta
- Hylla
  Långtidsförvaring
- Jag
  Utför
  arbetsuppgifter

Mitt Kontor

Hardware

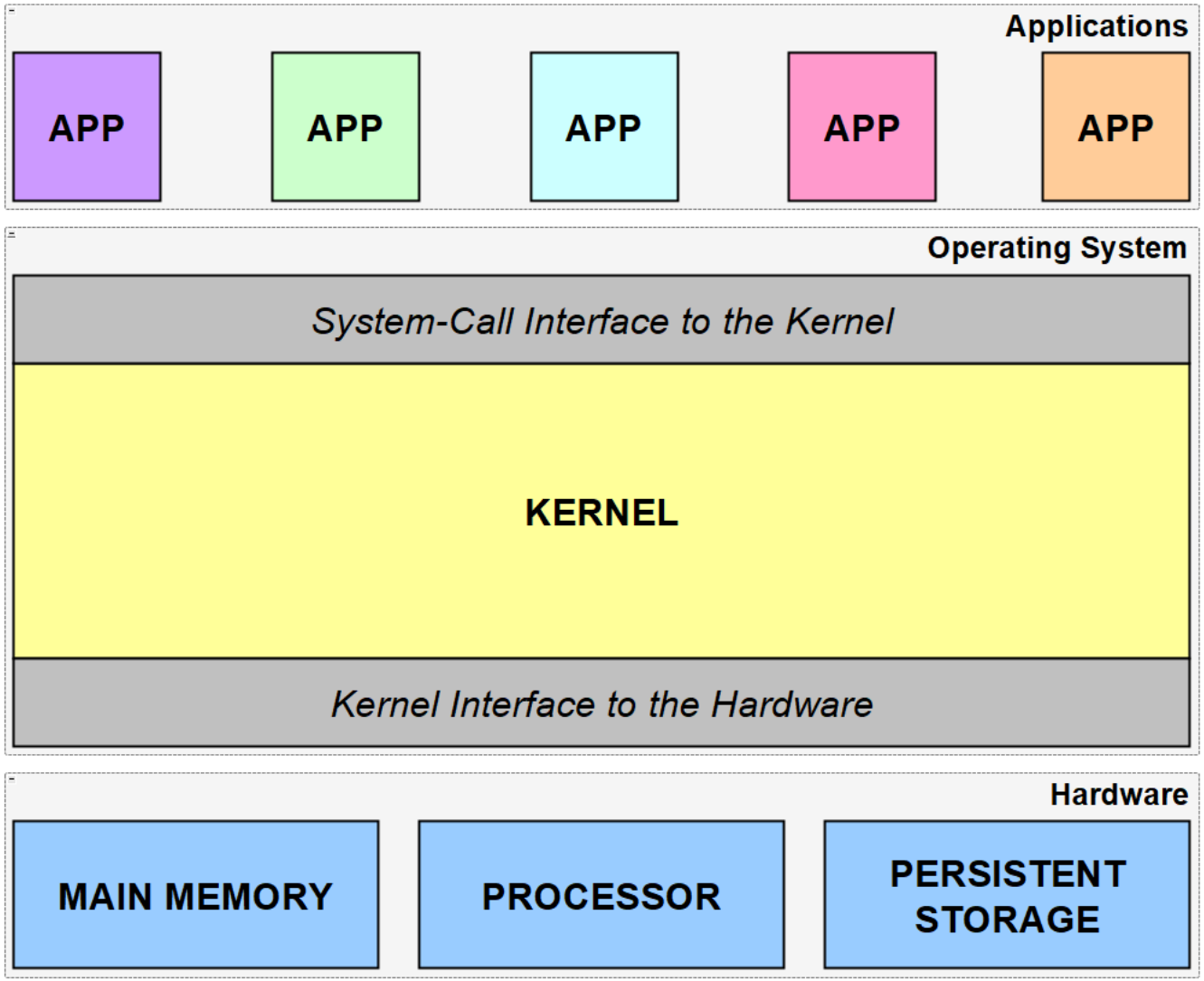
Hårdvara

En Liknelse…

- **Primärminne**
  Arbetsyta
- **Sekundärminne**
  Långtidsförvaring
- **Processor**
  Utför arbetsuppgifter

01_Ideapad_530S_Hero_Presentation_mode_OnyxBlack_B_NG, by BTNHD Production, 2018, Flickr (http://flic.kr/p/25DqcK2). CC BY-NC-ND 2.0.

Konceptuell dator



Till nästa gång

- Läs kurslitteraturen
- Se video-demonstrationer
- Onsdag: föreläsning, *Operativsystem*
- Fredag: laboration

Tack för idag